RAJALAKSHMI ENGINEERING COLLEGE

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D2

Question 1 | Correct Marked out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
    {
         i++;
         s += i;
     }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main()
{
    int n;
    scanf("%d", &n);

    int i = 1;
    int s = 1;
    int count = 3;

    while (s <= n)
    {
        count += 3;
        i++;
        s += i;
    }

    printf("%d", count);

    return 0;
}
```

Check

	Input	Expected	Got	
✔	9	12	12	✔
✔	4	9	9	✔

Passed all tests! ✔

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